

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 June 2024

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to June 2024 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. A molecule of carbon dioxide contains [1]

- A) covalent bonds
- B) ionic bonds
- C) no bonds
- D) metallic bonds

2. Electrolysis requires [1]

- A) pure water with no ions, always
- B) a molten or aqueous ionic compound
- C) a solid ionic compound with no mobile ions
- D) a sugar solution

3. The volume of a liquid is measured with a [1]

- A) metre rule
- B) stopwatch
- C) measuring cylinder
- D) beam balance only

4. Metals are good thermal conductors because they have [1]

- A) free electrons that transfer energy
- B) only covalent bonds
- C) a vacuum inside
- D) no particles

5. Which is correct? [1]

- A) silver displaces sodium
- B) copper displaces magnesium from $\text{MgSO}_4(\text{aq})$
- C) magnesium displaces copper from $\text{CuSO}_4(\text{aq})$
- D) gold displaces zinc

6. Ionic bonding is the [1]

- A) transfer of electrons from metal to non-metal, forming ions
- B) hydrogen bonding in water only
- C) attraction between nuclei only in metals
- D) sharing of a pair of electrons

7. When a metal is oxidised, [1]

- A) the metal always melts
- B) pH becomes 7 always
- C) oxygen is removed
- D) oxygen is added / electrons are lost

8. Acid + carbonate produces [1]

- A) water only
- B) salt + hydrogen
- C) salt + oxygen
- D) salt + water + carbon dioxide

9. A plant cell in a concentrated sugar solution may become [1]

- A) plasmolysed
- B) photosynthetic
- C) turgid
- D) a gamete

10. A high temperature denatures an enzyme because it [1]

- A) increases pH only
- B) turns it into starch
- C) changes the shape of the active site
- D) adds a coenzyme

11. Which energy conversion is correct? [1]

- A) sound → nuclear always
- B) gravitational potential → kinetic as an object falls
- C) chemical → kinetic in a battery at rest
- D) light → mass always

12. Which organelle is the site of photosynthesis? [1]

- A) mitochondrion
- B) nucleus
- C) ribosome
- D) chloroplast

13. The image in a plane mirror is [1]

- A) virtual, upright and laterally inverted
- B) real and magnified always
- C) real and inverted
- D) on the surface of the mirror only

14. Pressure is defined as [1]

- A) A / F
- B) $F \times A$
- C) $F + A$
- D) F / A

15. Carbon dioxide in air is tested with [1]

- A) a lighted splint
- B) iodine
- C) blue litmus only
- D) limewater, which turns milky

16. Anaerobic respiration in yeast produces [1]

- A) lactic acid only
- B) oxygen
- C) starch
- D) ethanol and carbon dioxide

17. A force of 20 N acts on an area of 5 m². The pressure is [1]

- A) 25 N/m²
- B) 4 N/m²
- C) 100 N/m²
- D) 5 N/m²

18. Lipase, with bile, digests fats to [1]

- A) amino acids
- B) starch
- C) maltose
- D) fatty acids and glycerol

19. Phloem tissue transports [1]

- A) sucrose and amino acids
- B) soil particles
- C) water only upwards
- D) oxygen

20. The word equation for photosynthesis is [1]

- A) nitrogen + water → glucose + oxygen
- B) glucose + carbon dioxide → oxygen + water
- C) glucose + oxygen → carbon dioxide + water
- D) carbon dioxide + water → glucose + oxygen

21. Pollination is [1]

- A) transfer of pollen from anther to stigma
- B) dispersal of seeds
- C) fusion of gametes
- D) germination of the seed

22. A unit of density is [1]

- A) N
- B) g/cm³
- C) W
- D) J

23. A lighted splint is used to test for hydrogen. A positive result is [1]

- A) a pop sound
- B) limewater turning milky
- C) a glowing splint relighting
- D) bleaching litmus

24. Neutral solutions have pH [1]

- A) 1
- B) 0
- C) 14
- D) 7

25. If the resultant force on an object is zero, the object **[1]**

- A) is at rest or moves at constant velocity
- B) must be falling
- C) must be accelerating
- D) must be at rest only

26. Ciliated cells in the trachea **[1]**

- A) make insulin
- B) absorb digested food
- C) sweep mucus and trapped particles
- D) pump blood

27. The products of digestion are absorbed mainly in the **[1]**

- A) oesophagus
- B) trachea
- C) bladder
- D) ileum (small intestine)

28. The halogens are in **[1]**

- A) Group 0
- B) Period 1 only
- C) Group 7 / 17
- D) Group 1

29. Which is true of magnets? **[1]**

- A) North-south attract
- B) Like poles attract
- C) South-south attract
- D) North-north attract

30. In double circulation, blood passes through the heart **[1]**

- A) only at night
- B) twice per circuit of the body
- C) not at all
- D) once per circuit of the body

31. The SI unit of speed is **[1]**

- A) kg
- B) J
- C) m/s
- D) N

32. Which is true of magnets? **[1]**

- A) North-south attract
- B) Like poles attract
- C) South-south attract
- D) North-north attract

33. Diffusion is fastest in a **[1]**

- A) crystal lattice at 0 K
- B) solid
- C) liquid
- D) gas

34. Osmosis is the movement of **[1]**

- A) water from a dilute solution to a more concentrated solution through a partially permeable membrane
- B) oxygen through the stomata only
- C) any solute from high to low concentration
- D) mineral ions against a concentration gradient only

35. Each enzyme works best at its **[1]**

- A) freezing point
- B) optimum pH and temperature
- C) boiling point
- D) ignition temperature

36. An object has mass 20 g and volume 10 cm³. Its density is **[1]**

- A) 200 g/cm³
- B) 2 g/cm³
- C) 30 g/cm³
- D) 10 g/cm³

37. The SI unit of energy is the **[1]**

- A) watt
- B) pascal
- C) joule
- D) newton

38. During electrolysis, cations move to the **[1]**

- A) cathode
- B) fuse
- C) anode
- D) voltmeter

39. Alkali metals (Group 1) become more reactive **[1]**

- A) up the group
- B) when they lose the nucleus
- C) down the group
- D) only in period 2

40. Which gas is produced during photosynthesis? **[1]**

- A) oxygen
- B) nitrogen
- C) hydrogen
- D) methane

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).